

Lemma. Given sequences $\{a_n\}$ and $\{b_n\}$, define $A_n = \sum_{k=0}^n a_k$, $B_n = \sum_{k=0}^n b_k$.

Then, for any $q \in \mathbb{N}$,

$$\sum_{n=0}^q a_n b_n = \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q = \sum_{n=0}^{q-1} \left(\sum_{k=0}^n a_k \right) (b_n - b_{n+1}) + b_q \cdot \sum_{n=0}^q a_n.$$

In particular, if $0 \leq p \leq q$,

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$$

Proof. Since $a_n = A_n - A_{n-1}$, ($A_{-1} := 0$)

$$\begin{aligned} \sum_{n=0}^q a_n b_n &= \sum_{n=0}^q (A_n - A_{n-1}) \cdot b_n = \sum_{n=0}^q A_n b_n - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + \sum_{n=0}^q A_n b_{n+1} - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + A_q b_{q+1} \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q \end{aligned}$$

Particular, $\sum_{n=p}^q a_n b_n = \sum_{n=0}^q a_n b_n - \sum_{n=0}^{p-1} a_n b_n$

$$\begin{aligned} &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - \left(\sum_{n=0}^{p-2} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} \right) \\ &= \sum_{n=p-1}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_{p-1} \\ &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + \underbrace{A_{p-1} b_{p-1} - A_{p-1} b_p + A_q b_q - A_{p-1} b_{p-1}}_{\text{Cancel}} \end{aligned}$$

Thm. If $\sum a_n$ Converges and $\{b_n\}$ is monotone bounded, then $\sum a_n b_n$ Converges.

Proof. Since $\{b_n\}$ is monotone bounded, it is convergent. put $b = \lim_{n \rightarrow \infty} b_n$, $A_n = \sum_{k=0}^n a_k$.

WLOG, suppose $\{b_n\}$ is monotone decreasing. Then, $\{b_n - b\}$ Converges to Zero, and decreasing.

Let $M > 0$ be a bound of A_n . Given $\varepsilon > 0$, put $N \in \mathbb{N}$ such that $n \geq N \Rightarrow |b_n - b| < \frac{\varepsilon}{2M}$.

By Abel's formula, for all $q \geq p \geq N$,

$$\begin{aligned} \left| \sum_{n=p}^q a_n (b_n - b) \right| &= \left| \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q (b_q - b) - A_p (b_p - b) \right| \\ &= M \cdot \left[\sum_{n=p}^{q-1} |b_n - b_{n+1}| + |b_q - b| + |b_p - b| \right] \\ &= M \cdot \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + (b_q - b) + (b_p - b) \right] \\ &= M \cdot [b_p - b_q + b_q - b + b_p - p] \\ &= 2M(b_p - b) < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon \end{aligned}$$

Therefore, by the Cauchy Criterion for Series, $\sum_{n=0}^{\infty} a_n (b_n - b)$ Converges.

Moreover, since $\sum_{n=0}^{\infty} a_n$ Converges, so is $b \cdot \sum_{n=0}^{\infty} a_n$. Finally,

$$\sum_{n=0}^{\infty} a_n (b_n - b) + \sum_{n=0}^{\infty} a_n \cdot b = \sum_{n=0}^{\infty} a_n b_n$$

Converges, as sum of two convergent series. $\square$

Note: If $b \neq 0$, then enough to Converges that $\sum a_n$ bounded, not necessary Converges,
 it is

